Churches, Religion and Development

A Broad Empirical Study from Russia

Abstract

This project is a broad study of Religion and Development in Russia. I create an instrument for the number of churches in the regions of Russia using exogenous variation in the deportation of clerus workers in the Soviet repressions of 1939. I find that churches apparently do not affect most aspects of development at the regional level, except perhaps for some specific criminal offenses. Two clear limitations of these results are the rather noisy regional-level data and the locality of the treatment effects of my instrument. Since this is pioneering work for Russia, I do not examine any mechanisms in detail, but rather broadly describe the whole picture in general.

1 Introduction

Religion has been a key aspect of human life for several thousand years. Throughout history, it might have also been a key aspect of societal development. There is evidence of a direct positive effect of religious practices on economic growth (Barro and McCleary, 2003). Potential channels include the buildup of trust (Blum and Dudley, 2001) and changing work ethics (e.g. Basten and Betz (2013)). At the same time, some specific practices like fasting could instead reduce productivity (Campante and Yanagizawa-Drott, 2015). Perhaps even more than through beliefs, religion might have affected growth indirectly, for example, through its effects on literacy (Becker and Woessmann, 2009).

More recently, literature has shifted its focus from studying history towards examining the current effects that religion might have on contemporaneous development or political outcomes. There are several interesting papers about Islamic practices, the Ramadan in particular (see e.g Michele's upcoming paper). Concerning Christianity, most studies focus on fertility (Berman et al. (2018), Sander (1992). For a more detailed overview of the field of Religion, Iyer (2016) is a good reference.

The Russian Orthodox Church has been receiving little attention from economists, and the goal of this project is to start filling this gap. The Russian setting provides an interesting institutional setup for research, as it might both still have a major impact on the economy and provide for several good sources of exogenous variation.

First of all, the Church in Russia appears to still be used for the political goals of the state. For instance, the Church might have made some rather strong statements regarding the sensitive topic of the conflict in Ukraine (Pillay, 2024). It appears that religious aspects sometimes might take second place behind political aspects in the preaches of leaders.

Second, the Russian Orthodox Church endured a vast amount of historical shocks. Russia is a multinational country, and the expansion of the empire might have had different effects on different locations in terms of religion. Moreover, there were several splits

happening in the XVI, XVII, and XX centuries. Additionally, religion in Russia was quite often affected by conflicts, as, for example, the Napoleonic Wars in the early XIX century. Most importantly, the Soviet Era forcefully relocated high-ranked religious figures to gulags and camps, which appears to be exogenous for the regions they were relocated to. Many popes and religious leaders were characterized as enemies of the state, who have been shown to bring a positive human capital effect to gulags (Toews and Vézina, 2025).

I collect micro-data for self-reported religiosity, as well as macro-data on the number of churches and various outcome variables. I first check (on a level of correlations) whether the number of churches in a region actually does represent the religiosity of individuals there. Next, construct several potential instruments for the number of churches, including historic church construction and deportation data. I find that the number of deported clerus workers in a region is strongly correlated with the number of churches today. This holds when controlling for the number of deported people in general, as well as the presence of a Gulag or camp in the region to begin with. I also argue that the exclusion restriction should hold for that instrument.

I examine the relationship between churches and different development outcomes, such as fertility, divorce and marriage rates, education, and crime. Simple correlations suggest that there is some negative interplay between churches and fertility, as well as some positive interplay between churches and abortions. However, by using the instrumental approach, I show that most of these relations disappear when looking for causal inference. I also show that having more churches in a region seems to reduce the number of criminal cases related to the illegal distribution of drugs.

The rest of the project is structured as follows. In Section 2 I describe data and compare micro and macro measurements of religiosity. In Section 3 I describe several approaches to instrumentation and show which turn out to be valid and which fail to be promising. In Section 4 I show and discuss my results. In Section 5 I conclude.

2 Data

I take data from three sources. Individual data on the religious affiliation of individuals comes from the Russia Longitudinal Monitoring Survey RLMS-HSE (2024). Data on churches comes from Temples.ru (2025) which I scrape myself using Selenium. Finally, data on fertility, marriages, crime, and education come from the regional socio-economic yearbook provided by Rosstat (2024) and digitalized by Tochno.st.

For reported religiosity I use the last wave (2024) of the RLMS-HSE (2024). This is the largest survey dataset conducted in Russia that comprises a set of questions ranging from labor participation, via education and migration, to religion, family, and other personal information. This is the only source of data known to me that covers self-perceived religious beliefs. At the same time, a huge disadvantage of the dataset is that the survey covers only specific locations, namely, 16 larger cities and 23 smaller conglomerations, a total of

39 locations throughout 33 regions. Six regions are represented twice; in each case, there is a representative larger city (region capital) and another smaller municipality. There is a total of 16,882 individual observations. The under-representation of regions makes it rather complicated to use historical shocks to cities and regions for micro-instrumentation.

The survey includes three main questions concerning religions. The first outcome of interest is the religion a person considers themselves to belong to (Q151). I create an 'Orthodox' variable that is equal to one if the person considers themselves an orthodox and that is equal to zero otherwise. Another outcome variable is what a person thinks about religion and supernatural powers in general (Q152). Here I create two variables. First, 'Believer' is equal to 1 if the person is a believer and 0 otherwise. Second, 'Any Believer' is equal to 1 if the person is either a believer or 'rather a believer than not' and 0 otherwise. Finally, I take data on a measure of whether a person visits divine services, meetings, or other religious events and if yes, how often (Q154). Here, there is a measure from 1 (never) to 7 (once a week or more often). I also create two dummies for those visiting religious services once a week and for those visiting at least once a month. For all dummies, I delete all answers of the 'does not know' and 'refuses to answer' type, rather than coding them as zeroes.

For data on churches, my main source is a website Temples.ru (2025) that contains data on 17,705 (working) churches in the regions of Russia. Each entry includes the name, the year of build, the exact location, a short description, and some other information. Importantly, there is a distinction between wooden and stone churches, as well as a categorization of churches into working, preserved but not working, not preserved, and those still under construction. By scraping this website with Selenium, I get data on both working and preserved churches in each region. Additionally, I find the number of wooden and stone churches that were built before 1917, the year of the Russian Revolution.

Importantly, the website of Temples.ru (2025) is created by enthusiasts, so I compare their data on the total number of churches with data from a more official source on Elitsy (2025). I find little discrepancies of about 1–2% in both directions between the two data sources, which suggests that Temples.ru (2025) might be slightly off the correct total number, but that the difference should be small. Notably, compared to Elitsy (2025), the website of Temples.ru (2025) overstates the number in the European parts of Russia, but usually slightly understates the number for regions further away from Moscow.

The difference might be observed due to different definitions of 'working' churches, as the usage might also vary significantly across observations. In particular, there are many villages that are small nowadays but experienced more prosperous times during the era of the Russian Empire. These villages sometimes have empty but very well-preserved churches, which are used once a year on specific church holidays. I do not address any concerns with attrition biases in my project mostly due to the matter of time constraints, but also because the discrepancies are rather small.

For outcomes, I use data from regional socio-economic yearbooks provided by Rosstat (2024) and digitalized by Tochno.st. I collect data on fertility and abortions, marriages and divorces, education, as well as data on the number of different criminal cases. For each outcome variable, I take data as of the year 2023, since this is the most recent available information (published in 2024).

Table 1: Descriptive Statistics

Variable	N	mean	sd
Number of Churches	39	270.7	215.6
Orthodox	13,948	0.819	0.385
Believer	13,634	0.263	0.440
Believer if Orthodox	11,134	0.255	0.436
Believer & Rather Believer than Not	13,634	0.790	0.407
Any Believer if Orthodox	11,134	0.870	0.336
Church Every Week if Orthodox	11,280	0.0099	0.0991
At Least Once a Month if Orthodox	11,280	0.0483	0.2144
Every Week if Orthodox Believer	2,822	0.0343	0.1822
Every Month if Orthodox Believer	2,822	0.137	0.345

For macro-data means are taken over regions.
For micro-data means are taken over individuals

Table 1 shows some descriptive statistics for my variables. The mean number of churches over 39 region entries is 270. Notably, when taken over individuals, the mean jumps to 369, which suggests that larger regions are represented more often in the survey. About 82% consider themselves orthodox, but only 25.5% of them are real believers. Moreover, less than 1% of orthodox people visit church every week, with this number increasing to about 3.4% for those orthodox who are believers.

Table 2: Outcomes. Descriptive Statistics

Variable	N	Mean	SD	Mean Log	SD Log
Total Fertility	80	1.427	0.269		
Abortions per Cent Births	80	43.84	14.32		
Marriages per Mill	80	6.23	1.32		
Divorces per Mill Marriages	80	852.9	456.3		
Murders	80	90.2	76.7	4.16	0.86
Stealings	80	357.3	317.6	5.36	1.26
Hooliganism	80	16.4	28.5	2.24	1.05
Bachelors per Mill	80	246.9	110.6		

Table 2 shows some descriptive statistics for the main outcomes. The average number of children per woman in a region is 1.43, while the average share of abortions per birth is about 43.8%. There are, on average, 6.23 marriages per 1000 people in a region, while, on average, 85.3% of marriages end up in divorce. Notably, the (log) number of criminal cases is reported without adjusting for population size.

In order to later use the number of churches as a main regressor, I first examine individual religious affiliation by considering correlations between the number of churches within a region and reported religiosity. I include some regional control variables such as log GDP, population, area, as well as a dummy for being a big city (defined as a regional capital), and Federal Area fixed effects (e.g. Central, Ural, Volgal). I also include some interaction terms between the log variables and the 'big city' variable. To account for regional correlation, I cluster the heteroskedasticity-robust errors at the regional level.

Table 3 reports the results. The regressions imply that there are strong correlations between the number of churches and religiosity. The association is about ten times smaller, but still different from zero, for individuals that live in big cities. For example, on average, an orthodox is expected to be about 22.3 percentage points more likely to be a believer if they live in a region with double the amount of churches and if they do not live in the city. For city-dwellers, the increase is only about 2.1 percentage points, which is still about 8.2% relative to the mean. All of these results are robust to deleting megapoli (Moscow, Moscow Region and St.Petersburg) as well as Muslim regions (Kabardino-Balkaria and Tatarstan in the RLMS) from the sample.

Table 3: Churches and Religiosity

		If Orthodox		
				If Believer
	Orthodox	Believer	Any Believer	Every Week
	(1)	(2)	(3)	(4)
Log Number of Churches	0.350***	0.223*	0.157***	−0.0102***
	(0.087)	(0.115)	(0.052)	(0.0032)
Log Number of Churches × Big City	−0.196**	−0.202**	−0.110	0.0040
	(0.087)	(0.088)	(0.071)	(0.0040)
Federal Area FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Number of Observations	13,709	8,657	8,657	9438
Mean of Dep. Variable	0.819	0.255	0.870	0.039
R-squared	0.184	0.027	0.024	0.013

Notes: The number of churches is taken for the region an individual lives in. A big city is defined as a regional capital. The outcome variables and controls are described above. Robust and region-level clustered standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We might suggest that there is some causal relationship with people being more likely to be religious if there is a religious object (and preacher) nearby. For individuals in a village, the corresponding number of churches within the region is related to the *probability* of having a church, while in a city it would rather be the *number* of churches in the city. The massive reduction in the coefficient implies that, given a causal relation, it is rather the intensive margin that is important rather than the extensive margin.

There is one fact that is not very consistent with my logic. The probability of visiting church every week is *negatively* related to the amount of churches in the region. There is an unreported similar result for the probability of visiting a church every month. The only potential (causal) explanation that I could give here is that there might be some selection in play. That is, in places where it is harder to get access to religious services, those who consider themselves religious are only those who indeed have very strong preferences towards religion and hence actually do comply with visiting the church even under scarce supply. However, there could potentially be any other explanation for this result, which calls for caution when interpreting coefficients below in Section 4.

The number of churches in a region might be the closest proxy variable to measuring the level of religiousness at the macro level, as, unfortunately, to my knowledge, there are no better surveys than that of the RLMS. However, the relationship between religiosity and the number of religious buildings might not be that simple, as churches often are built to live for centuries. On one hand, more religious people supposedly build more churches. On the other hand, people with easier access to religious services could be more likely to consider themselves as religious. More generally, one could think of alternative explanations, as, for example, richer people could be more likely to build churches and simultaneously more likely to be religious. Hence, it is better to understand the results from Section 4 as the effect of churches themselves rather than an effect of religiosity.

3 Instrument Construction

I construct and test for validity two instruments. The first instrument is the amount of clerus working people deported to a gulag or camp in a receiving region. The second instrument is the share of stone churches among those constructed prior to 1917. The former turns out to be relevant, while the latter fails to have a strong enough F-statistic. I provide a more detailed description of both below.

As I am very inspired by Gerhard's paper on enemies of the people, I try to apply his data to instrumenting the amount of churches in Russian regions today. The idea is that some relocated popes might have started preaching in the regions they were transferred to and thereby affected religiosity. There is some anecdotal evidence supporting this mechanism, as, for example, Tobolsk, a minor city some Orthodox leaders were transferred to, has a significantly higher per capita amount of churches than the larger close-by city of Tyumen, the capital of the region. At the same time, the choice of a specific camp

for each cleric appears to be random when controlling for whether there is a gulag in the region to begin with, as well as controlling for the total number of inmates.

Technically, I assume that convicts were sent to camps randomly, that is, the choice of a gulag for each person was not affected by the fact whether they were a clerus worker or not. I cannot test this assumption directly. A good thing to do would be to take pre-gulag data on regions and look whether it correlates with clerus inmates. However, I do not have access to such data, particularly because Soviet regions do not always coincide with the current ones. One concern with the exclusion restriction might be that clerus workers are most probably older than the average inmate, which could result in them systematically being sent to work on less physically demanding tasks (into different regions).

I take data on camps from the replication package of Toews and Vézina (2025) and conduct a first-stage regression. I control for region (log) population, GDP, area, the unemployment rate, a dummy for being a metropolitan city, and for Federal Area fixed effects. Importantly, I control for a dummy that indicates whether there was a gulag or other camp in the region in 1939, as well as for the log number of inmates at that time.

Table 4: Effects of Clerus Inmates on Churches

	(1)	(2)
	Log Number of Churches	Log Number of Churches
Log Number of Clerus Inmates	0.939	0.539***
	(0.666)	(0.196)
Log Number of State Enemies	−0.998	−0.666**
	(0.719)	(0.256)
Presence of Gulag	4.846	3.706**
	(3.905)	(1.615)
Controls	Yes	Yes
Federal Region FE	No	Yes
Observations	80	80
Mean of Dep. Variable	4.30	4.30
Adj. R-squared	0.02	0.82

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 shows the results of the first-stage regression. The log number of clerus inmates has a positive effect on the log number of churches. The estimates are very imprecise without Federal Area fixed effects, but become statistically significant at the 1% level when adding the latter. The estimates suggest that a 1% increase in the number of clerus inmates leads to a 0.54% increase in the number of churches. This effect appears to be economically quite large. The F-statistic is about 9, which is slightly below the recommended value for instrumental usage.

Alternative Instrument

As this is a general study, my goal is not to find one single instrument and use it, but to explore as many ideas as possible. Instruments for religiosity might prove useful for projects and research later on.

In another approach, I look at variation in church building material prior to 1917 and try to exploit differences in survivability for instrumenting the number of churches nowadays. I do not find this instrument to be relevant, so if you want, you can skip the rest of this Section. I decided to include this part because I still see some value and potential for further research in this direction.

The choice between wood and brick building materials appears to not reflect the extent of the religiosity of the builder (or sponsor) of the church. Hence, it should also not reflect the historic religiosity of a region or village. Rather, it is quasi-exogenous with respect to some controls. Obviously, it correlates with the wealth of a village at the time of construction, as well as it does with the infrastructure around it. Additionally, it correlates with the time of construction itself, which might in turn correlate with the religiosity of a region. However, controlling for that, the building material should be exogenous.

At the same time, wooden churches most likely deteriorate faster than brick churches. Moreover, during the Soviet times, most churches were not receiving any care or repairs, and were not restored if they became ruined. Hence, 100 years later, regions with brick churches would still have access to them, while regions with wooden ones would more likely have lost them. This gives a nice exogenous variation in access to religious services.

Table 5: Historic and Current Number of Churches

	Log Total Number of Churches in 2025		
	(1)	(2)	(3)
Log Number of Stone Churches 1917	0.443***	0.009	
	(0.086)	(0.125)	
Log Number of Wooden Churches 1917	0.179**		0.008
	(0.078)		(0.084)
Log Total Number of Churches 1917		0.594***	0.596***
		(0.142)	(0.089)
Controls	Yes	Yes	Yes
Federal Region FE	Yes	Yes	Yes
Observations	80	80	80
Mean of Dep. Variable	4.300	4.300	4.300
Adj. R-squared	0.929	0.937	0.937

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 reports the results of a first-stage regression regarding historic building material. Churches built out of brick appear to survive better than those built out of wood, with the coefficient almost three times as high (0.44 vs 0.18). However, each type of material appears not to play any role when controlling for the total number of churches built prior to 1917. Not reported here, I get similar results when I estimate the effect of the shares of stone or wooden churches.

A third option that I could exploit is the War with France in 1812. As it was disproportionately large and impactful in Europe and Russia in particular, many high-ranked noblemen took part in the fighting, often personally. In some families at that time, surviving a war was often attributed to God's mercy, requiring praise. Technically, I might be able to get some data on noblemen that would be rich enough to build a church. Moreover, with Russian archive data, it should be possible to identify those who participated and those who fell. I might expect that the deaths of noblemen could provide some natural variation to the probability of church construction nearby the corresponding noble residences. This would be another, perhaps even more solid, idea for an instrument. At the same time, working with archive data reaches far beyond the scope of a project, so I leave that idea for further research.

Another potential extension for future research regarding my approach would be a further granulation of the data from region levels to smaller units like municipalities or even villages within one region. This is potentially feasible, as all churches in the Temples.ru (2025) database are geocoded. Such an extension would allow the elimination of the significant amount of noise that comes from aggregation. However, once again, this reaches beyond the limits of a single project.

4 Results

In what follows, I describe the results and compare OLS estimation to my IV strategy. Tables 6, 8, and 10 report simple regression results for fertility and abortions, marriages and divorces, and six different crime rates. Tables 7, 9, and 11 report the same estimates from an instrumental approach. Table 12 reports both OLS and IV estimates for the relation between churches and education. I discuss the results altogether, as, once again, I try to look at the whole picture instead of focusing on one specific outcome.

The ordinary least squares estimates provide a surprising correlation. The amount of churches is *negatively* correlated with fertility and *positively* with abortions. This appears to go against the position of the Orthodox Church, which encourages fertility and strictly condemns abortions. The correlation between churches and marriages is positive and marginally significant, while the correlation between churches and divorces per thousand marriages is negative and also marginally significant. For criminal records, most estimates are insignificant, with the exception of stealing convictions that are, for some reason, *positively* correlated with churches. Finally, neither the number of bachelor

graduates nor the number of educated specialists shows any correlation.

Table 6: OLS of Fertility on the Number of Churches

	(1) Tot. Fertility	(2) Fertility/Mill	(3) Abortions/Cent Births	(4) Abortions/Woman
Log Churches	−0.116*	−1.122**	3.932***	0.511
	(0.062)	(0.465)	(1.403)	(0.686)
Mean Dep. Var	1.427	8.694	43.839	16.305
Federal Region FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	80	80	80	80
Adj. R-squared	0.409	0.627	0.627	0.633

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: IV Estimates Churches and Fertility

	(1) Tot. Fertility	(2) Fertility/Mill	(3) Abortions/Cent Births	(4) Abortions/Woman
Log Churches	−0.208	−0.608	−6.363	−3.928
	(0.167)	(0.923)	(7.418)	(3.398)
Log Enemies	−0.070*	−0.399*	0.342	−0.454
	(0.040)	(0.231)	(2.379)	(1.023)
Mean Dep. Var	1.427	8.694	43.839	16.305
Federal Region FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	80	80	80	80
Adj. R-squared	0.358	0.600	0.392	0.372

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: OLS of Mariage on the Number of Churches

	(1) Marriages/Cent	(2) Divorces/Cent	(3) Divorces/Mill Marriages
Log Number of Churches	0.252*	0.027	−102.689*
	(0.141)	(0.140)	(54.885)
Mean of Dep. Variable	6.234	4.887	852.925
Federal Region FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	80	80	80
Adj. R-squared	0.662	0.462	0.850

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: IV Estimates Churches and Marriages

	(1) Marriages/Cent	(2) Divorces/Cent	(3) Divorces/Mill Marriages
Log Churches	0.136	−0.470*	−106.913
	(0.534)	(0.271)	(70.327)
Log Enemies	0.068	−0.225***	−37.431*
	(0.151)	(0.076)	(19.439)
Mean of Dep. Variable	6.234	4.887	852.925
Federal Region FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	80	80	80
Adj. R-squared	0.657	0.319	0.847

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: OLS of Criminal Records on the Number of Churches

	Log Number of Cases in 2023					
	Murder	Stealing	Selling Drugs	Brigandage	Harm Infliction	Hooliganism
	(1)	(2)	(3)	(4)	(5)	(6)
Log Churches	−0.004	0.372**	0.147	0.256	0.250	−0.029
	(0.080)	(0.173)	(0.091)	(0.158)	(0.163)	(0.129)
Mean Dep. Var.	4.161	5.363	7.194	3.318	4.976	2.238
Federal Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80	80	80	80	80	76
Adj. R-squared	0.854	0.843	0.859	0.828	0.823	0.618

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: IV Estimates Churches and Criminal Records

	Log Number of Cases in 2023					
	Murder	Stealing	Selling Drugs	Brigandage	Harm Infliction	Hooliganism
	(1)	(2)	(3)	(4)	(5)	(6)
Log Churches	−0.326	0.198	−0.798***	−0.127	−0.011	0.040
	(0.269)	(0.309)	(0.284)	(0.200)	(0.258)	(0.487)
Log Enemies	0.013	0.075	−0.221**	−0.022	−0.046	−0.336**
	(0.098)	(0.085)	(0.103)	(0.066)	(0.074)	(0.163)
Mean Dep. Var.	4.161	5.363	7.194	3.318	4.976	2.238
Federal Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80	80	80	80	80	76
Adj. R-squared	0.787	0.834	0.463	0.779	0.785	0.631

Notes: The controls include log population, GDP, and area; the unemployment rate, a dummy for being a metropolitan city (Moscow or St.Petersburg), and a constant. Robust standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Churches and Education

	Num. Bachelors per Mill		Num. Specialists per Mill	
	(1)	(2)	(3)	(4)
	OLS	IV	OLS	IV
Log Churches	−12.384	70.122	3.375	31.644*
	(14.129)	(80.498)	(4.858)	(18.132)
Mean Dep. Var.	246.925	246.925	197.617	197.617
Federal Region FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	80	80	80	80
Adj. R-squared	0.248	−0.013	0.172	−0.124

Notes: Columns (1) and (3) report OLS estimates. Columns (2) and (4) report IV estimates. The controls include log population, GDP, and area; the unemployment rate, and a dummy for being a metropolitan city (Moscow or St.Petersburg). Robust and region-level clustered standard errors are reported in parentheses. The p-values are reported as * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The IV estimation shows a different picture. Most coefficients from above become insignificant, or even reverse signs, with a few exceptions. First, there is a marginally significant negative effect of churches on divorces. This might represent a Church policy of discouraging divorces, but it could also simply be a chance finding. Similarly, the number of graduated specialists is positively affected by churches at a marginally significant level.

Interestingly, for crime rates, additional churches within a region appear to significantly reduce the number of cases related to selling and distributing drugs. The coefficient is -0.798, which implies that a 1% increase in the number of churches reduces the number of such convictions by about 0.8%. This is quite a large effect also in economic terms. It is not clear, however, why other records are not affected by churches. By looking at the coefficients, it appears that, perhaps, murders, for example, might have also been found to be affected, but have too low of a variation to be identified in a setting with 80 observations. From my results, I suggest that studying the relation between religiosity and crime in Russia might be the most promising prospect for future research.

I attribute the extinction of the OLS coefficients in the instrumental approach to reverse causality. That is, I suspect that regions with specific human preferences and behavior are more likely to have a church, rather than that churches and priests actually affect said human behavior.

Note also that the criminal records show only disclosed crime, so there might also be some attrition bias in play here. For example, building an additional church in a region might distort money from hiring more policemen. Hence, the results from the crime section should also be addressed carefully.

I also explicitly point out the fact that I only have 80 regions to observe. And it might

be very likely that this is the only reason why I do not find many effects. I cannot solve this issue within the scope of my project, but there might be some potential for further extension here with a better granulation of the data.

Most of my analysis could be granulated to smaller units, as, once again, for churches, the exact location is available and technically could be scraped from the websites. Similarly, some data in Russia is published at the municipality level, so it might be possible to get outcome data at a smaller level as well. Finally, data on gulags and camps is very nicely geocoded in Toews and Vézina (2025), so this also allows for smaller granulation. The only problem that might occur here is that I am not too sure whether criminal statistics and court decisions, which looked most promising above, are publicly available at the municipality level or even at the level of courts.

Finally, it is very important to point out that my instrument affects a very specific subset of regions. Camps and gulags were usually located in distinct places in the North, Far East, and Central-Asian part of the Soviet Union. These regions definitely differ from Central-European parts of Russia, where most of the population is concentrated. Moreover, the analysis makes very little sense for the Muslim regions of Russia. In regressions not reported here, I find support for the fact that the instrumentally affected regions differ significantly from unaffected in terms of GDP and population. Obviously, I control for Federal Area fixed effects as well as for the simple fact of a region having a camp to begin with. Still, my results of the IV estimation can *only* be interpreted as local treatment effects and cannot be generalized to usual average treatment effects.

5 Conclusion

To sum up, I conduct a broad and rather descriptive study of churches and religiosity in Russia. The study includes addressing general concerns with the measurement of religiosity in regions, as well as some preliminary examination of relationships between religiosity, or more precisely, churches, and several developmental outcomes.

Using micro-data from the Russia Longitudinal Monitoring Survey, I compare measures of church presence to self-reported religiosity. I find a significant correlation between churches and reports of an Orthodox affiliation, as well as with reports of being a believer for those who are Orthodox. At the same time, I find that the number of churches is *negatively* related to actually visiting churches or attending other religious services. I attribute this result to selection effects, but I do not provide any proof of this mechanism at this stage, nor do I even examine this issue in more detail. Finally, I find that the churches in a region are more important for people in rural areas, which suggests that it is the intensive margin rather than the extensive margin that carries most of the effect. In other words, having at least one church in a village is more important than having one additional church in a city.

Next, I match data on churches with several regional developmental outcomes, and

look at relationships in this direction. OLS results show many different effects, but most of them disappear when using the instrumental approach. I suggest that this means that there is some reverse causality, so that regions that differ initially build more churches rather than that churches shape the different regions into what they are.

Importantly, I find a negative effect of churches on criminal cases related to selling and distributing drugs. My results suggest that 1% of additional churches in a region could lead to a 0.8% decrease in drug-related cases. This could be attributed to some disciplining effect of churches, but could also represent differences in money spent on religion as opposed to other public goods, such as law enforcement. Moreover, this coefficient can only be interpreted as a local average treatment effect rather than a global one. I also find a marginal negative effect on divorces and a marginal positive effect on specialist graduates, but I suppose these could be chance findings.

The main contribution of this project is that I construct an instrument for the number of churches in the regions of Russia. I find that deported clerus workers are relevant for the recipient region and argue that they are also exogenous when controlling for gulag characteristics. I claim that this might be fruitful for extensions. At the same time, I fail to find relevance in a different approach using church building material prior to 1917.

The main prospect for future research appears to be a smaller granulation of the data with the potential to use the instrument with more precision. The main outcome that seems to be promising to be studied in relation to religion is crime rates. Additionally, by using archive data on the Napoleonic Wars, it might be possible to create another instrument using exogenous variation in nobleman death rates across villages. That instrument would allow for addressing the locality concern related to gulag placement, as it would be more relevant for the European parts of Russia.

Note on AI usage

Overleaf-inbuilt AI helped me out with the English and the structure of the text. ChatGPT was my main companion while I was scraping the websites for data. Data-work in Stata was quite straightforward, so I actually did it myself this time. I asked various sources of AI for help with literature at different stages of the project.

Note on Self-Plagiarism

I should explicitly point out that my current project for the Political Economy course also touches on Religion. That project examines transitions of religious leaders and voting outcomes in the regions of the Russian Federation. The data sources, data collection, data analysis, and general writing work differ entirely for the two courses. However, there definitely are some intersections in the literature work as well as a common institutional setting I explore. This means I might have had to read fewer papers than my average classmate. If needed, I am ready to share my other project for further inspection.

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